

Harmonic function

1. Show that $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
2. Show that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
3. Show that $u = x^3 + 6x^2y - 3xy^2 - 2y^3$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
4. Show that $u = e^x(x\cos y - y\sin y)$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
5. Show that $u = x^2 - y^2 - 2xy - 2x + 3y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
6. Show that $u = 2x - x^3 + 3xy^2$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
7. Show that $u = e^x \cos y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
8. Can a constant function be harmonic? Why?
9. Is $u = x^2 + y^2$ harmonic? Why or why not?
10. Is $u = x^2 - y^2$ harmonic? What is its harmonic conjugate?
11. Show that $u = x^2 - y^2 - 4x + 6y$ is a harmonic function and hence find its harmonic conjugate v if $f(z) = u + iv$ is analytic.
12. Determine whether $u = 2x^2 - 4y^2 - 5xy - 4x + y$ is harmonic. If it is harmonic, find its harmonic conjugate $v(x, y)$.